Decision Analytics Journal 9 (2023) 100344

Contents lists available at ScienceDirect

Decision Analytics Journal

journal homepage: www.elsevier.com/locate/dajour

A comprehensive review of blockchain technology: Underlying principles and historical background with future challenges

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ARTICLE INFO

Keywords:
Blockchain technology
Consensus algorithms
Decentralized distributed ledger
Immutability
Cryptography
Transparency

ABSTRACT

Blockchain is a distributed digital ledger technology that has revolutionized businesses, industries, and commerce by eliminating the need for a central storage and control authority. Blockchain presents time-stamped and immutable blocks of data that are not owned by any single entity but rather managed by a group of nodes or computers where each block is secured and linked using cryptographic principles. The immutable and decentralized nature of blockchain has redefined trust, ownership, identity, and financial systems by providing a secure, fast, transparent, and pseudo-anonymous solution. This paper provides a comprehensive review of blockchain technology focusing on the historical background, underlying principles, and the sudden rise in the popularity of blockchain technology. The paper also discusses the various consensus algorithms of blockchain technology. Next, the paper focuses on the various application areas and prospective use cases of blockchain technology with the underlying challenges and issues. Further, the paper presents some unconventional use cases of blockchain technology. The study also reviews state-of-the-art articles to provide a comprehensive overview of the various aspects of blockchain technology in varied domains. The comparison between traditional database systems and blockchain technology is presented, and the appropriate scenarios where blockchain-based solutions may or may not provide the best solutions are also discussed. Further, it discusses some of the most infamous security breaches that impacted the blockchain industry in the recent past.

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https://doi.org/10.1016/j.dajour.2023.100344
Received 20 March 2023; Received in revised form 17 October 2023; Accepted 18 October 2023
Available online 20 October 2023
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1. Introduction

Blockchain technology has revolutionized the digital transformation of contracts, transactions, and records which form the defining structure of the economic, political, social, and legal systems that govern this world. The first generation of blockchain technology started with the introduction of bitcoin by the pseudonym Satoshi Nakamoto in the year 2008 [1], which introduced the concept of blockchain and marked the deployment of cryptocurrencies in financial applications involving cash, like digital payment systems. The second-generation blockchain was marked with the introduction of smart contracts that provided functionality beyond cash transactions like stocks, loans, and smart property. The third-generation blockchain had a broader application area that introduced blockchain-based solutions for applications beyond banking and finance and provides services in the areas of healthcare, government, science, etc. In less than a decade, blockchain technology has already seen three generations and is paving the way to enter the fourth generation with Artificial Intelligence (AI) and digital intelligence. The sudden popularity and growth of the technology can be estimated from the fact that in the year 2010 Laszlo Hanyecz, ordered two Papa John's pizzas worth approximately 30$ for 10,000 bitcoins. In the year 2018, the same bitcoins were estimated to be over $80 million [2]. This shows the exponential growth in technology and cryptocurrencies.

Blockchain has the potential to digitally transform the way the world transacts by allowing the contracts to be digitally embedded in databases that are transparent and provide security against tampering. This key attribute of blockchain technology eliminates the need for third parties and intermediaries like banks, lawyers, etc. Although the potential of blockchain is immense, there are some issues and challenges that need to be resolved for its widespread adoption.

1.1. Brief timeline of blockchain and related technologies

Blockchain is a digital distributed ledger that secures and links the digital records called "blocks" using cryptographic techniques. Although the term blockchain gained popularity in the year 2008 with the introduction of Bitcoin cryptocurrency [1], its underlying principles and concepts have been in use since the 1980s. In the year 1983, David Chaum proposed the concept of blind signatures [3], a new form of cryptography that allowed the automation of payments and constrained third parties from identifying the payee, amount, and time of payment. Later, in the year 1991, the concept of timestamped documents was introduced by a group of researchers who were trying to secure digital documents by stamping the documents with the date [4]. In 2004, Harold Thomas Finney II (Hal Finney) introduced the Reusable Proof of Work (RPoW) [5]. It was in 2008 when blockchain made its public debut with a paper titled "Bitcoin: A Peer-to-Peer Electronic Cash System" outlining the development of a secure and transparent currency without the involvement of a bank or central party [1]. The first bitcoin transaction took place in the year 2009 when a transaction of 10 BTC (BitCoin) was done to his friend Hal Finney [6]. This marked the beginning of the first generation of blockchain technology. In the year 2010, the world's first bitcoin exchange was set up by the name "Bitcoin market" [7]. In 2013, Vitalik Buterin proposed the concept of smart contracts in his white paper "Ethereum: The Ultimate Smart Contract and Decentralized Application Platform" [8]. In 2015 Ethereum launched its blockchain [9]. The Linux Foundation launched Hyperledger the same year which provides a set of tools for

Fig. 1. Key events in blockchain technology [6–14].

the creation of blockchain [10]. In 2016 the DAO project was built on top of Ethereum which raised US$ 150 million [11]. Later, in the same year, it was attacked causing a loss of around US$ 50 million [12]. Fig. 1 presents the major events in the development and widespread popularity of blockchain technology [6–14].

Ever since its inception in the year 2008, blockchain technology has gained attention from across the globe. Today, many nations are adopting this technology to bring reforms across various sectors including finance, healthcare, education, governance, supply-chain, agriculture, and energy. This paper presents the state-of-the-art of blockchain technology and gives a comprehensive overview of its historical background and evolution journey. The paper also presents in detail the various aspects of blockchain technology, its application areas, and the challenges faced.

The manuscript is organized into eight sections. To begin with, the paper sets the stage for the reader to understand the historical background and the timeline of events resulting in the popularity of blockchain technology. Section 2, provides the details of the past research and studies conducted in the field of blockchain technology. Section 3, presents a general introduction to blockchain technology including the structure of blocks and the basic working mechanism of a blockchain. Section 4, highlights the underlying concepts of the technology focusing on the key attributes of blockchain. The section also presents an analysis of the various consensus algorithms. The taxonomy and generations of blockchain are discussed in Section 5. The section further provides a comparison between the traditional databases and blockchain-based on multiple characteristics. Section 6 presents the use-cases of blockchain technology. It also provides a discussion on passwordless authentication using blockchain technology. The section also discusses the appropriate scenario for the implementation of blockchain technology over traditional systems. The various challenges and issues related to the adoption of blockchain technology are discussed in Section 7. The section also discusses the security threats and their root causes. Lastly, Section 8 presents the conclusion and future scope.

1.2. Major contributions

The presented work provides a comprehensive analysis and survey of blockchain technology covering all aspects from the background, history, needs, roles, issues, applications, and challenges. The major contributions of the paper are as follows

- **Highlighting the key challenges and issues in the large-scale adoption of blockchain technology**
  An exhaustive discussion on the issues and challenges associated with the mass adoption of blockchain is presented covering the various technical, social, environmental, and economic aspects.
- **Discusses the consensus algorithms used in blockchain technology in various application areas.**
  The major consensus mechanisms are identified and their respective pros and cons are highlighted.
- **An extensive discussion about the key application areas of blockchain technology is presented focusing on the current state-of-the-art.**
  A detailed discussion on the state-of-the-art blockchain implementations across multiple domains including decision analytics, healthcare, data management etc.
- **Highlights the application of blockchain technology in passwordless authentication**
- **Discussion about appropriate use-cases and situations where blockchains should be avoided.**
  Even though blockchain technology is finding its applications in varied domains, there are several situations and applications where blockchain technology is not preferred. In this paper, we have identified these use-cases where the use of blockchain technology must be avoided. To the best of our knowledge, these kinds of use-cases are not discussed in any of the related articles.

1.3. Methodology

A systematic literature review was conducted to retrieve the most relevant state-of-the-art in blockchain technology as per PRISMA 2020. Initially, the search query was run on ScienceDirect and IEEE Xplore